SUYASH ROY

Los Angeles, California
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EDUCATION

University of Southern California (USC) Jan 2025 – Expected Dec 2026
Master of Science in Computer Science CGPA: 3.71/4

Kalinga Institute of Industrial Technology (KIIT) Jun 2018 – Jun 2022
Bachelor of Technology in Information Technology CGPA: 9.19/10

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Java, TypeScript, JavaScript, C++
- **Machine Learning:** PyTorch, TensorFlow, Hugging Face, Scikit-learn, XGBoost, Deep Learning, NLP, Vision-Language Models, Post-training (LoRA/PEFT), Reinforcement Learning (coursework)
- **Recommendation & Retrieval:** Recommendation Systems, Ranking & Personalization, Vector Similarity Search (FAISS), BGE Embeddings, RAG, Feature Engineering, Propensity Modeling, Model Evaluation
- **Data & Deployment:** PostgreSQL, PostGIS, MongoDB, Redis, Pandas, NumPy, Docker, CI/CD, AWS, Vercel, Git, SLURM GPU Cluster

PROFESSIONAL EXPERIENCE

AI Implementation & Strategy Intern Jun 2026 – Present
Brasa Capital Management, Los Angeles, CA

- Lead a 2-person team building BrasaSuite, four AI-enabled applications for 20–25 investment professionals; own delivery end to end, from data exploration to production deployment, iterating with daily users.
- Deployed a human-in-the-loop LLM document pipeline that ingested 100K+ SharePoint files and now processes 6K/week into structured PostgreSQL/PostGIS data for investment analysis.
- Designed schemas for versioned investment metrics and geospatial analytics; own monitoring/observability, code reviews, go-live testing, and on-call for production tools.
- Built a live-testing OneNote MCP server, scraper, and dashboard that give nontechnical users self-serve data discovery, secured with Microsoft Entra ID.

Research Assistant Apr 2026 – Present
University of Southern California, Los Angeles, CA

- Fine-tuned CXR-LLaVA with LoRA in PyTorch on 10K image-report pairs and ran batch inference over 5,924 test images on USC's SLURM GPU cluster with zero errors.
- Built a claim-level evaluation framework attributing hallucinations across 5 evidence sources with RadGraph extraction and token likelihoods; synthesized 50+ papers.

Associate Software Developer (AI / ML) Jan 2022 – Dec 2024
Bain Capability Network, Gurugram, India

- Deployed ML regression models (Random Forest, XGBoost) on geospatial and census datasets covering 6M+ households, reducing RMSE by 18% to predict high-IRR deployment markets.
- Created broadband subscriber churn propensity scoring model 'LTS', used across 5+ strategic market expansion and acquisition cases.
- Formulated churn-prediction feature pipelines across 10+ global cases and Python batch pipelines analyzing 6M+ telecom records, optimizing memory usage and runtime.
- Modeled geospatial demand estimation using spatial joins over 5K+ telecom towers and 100K+ mobility data points.

Full Stack Developer Intern, CFO Tech May 2021 – Dec 2021
HighRadius, Bhubaneswar, India

- Devised REST APIs using Java Spring/Hibernate and ReactJS components for enterprise SaaS workflows; served as peer tester and authored a root-cause-analysis (RCA) checklist.

ACADEMIC PROJECTS

Lead Coder, CareerAgent, Los Angeles, CA May 2026 – Present

- Built a job recommendation engine that ranks openings by relevance to a candidate's resume, combining 1024-dimensional BGE embeddings, FAISS vector similarity search, and deterministic keyword-match scoring.
- Shipped four agents for discovery, semantic matching, resume tailoring, and real-time voice interviews, with LangGraph/Playwright fallbacks and Llama 3.3 critique.

NetConnectAI, Los Angeles, CA Jan – Feb 2026

- Built a RAG-based broadband recommender (React, Express) that geocodes a user's address to its census block and queries 3.6M FCC availability records across 389 providers in PostgreSQL within about 700 ms.
- Personalized plan recommendations with a Gemini chatbot grounded in user requirements, provider data, and Google Places reviews; built the Python/Pandas ETL rollup and a MongoDB cache with 4-tier fallbacks.

Lead Coder, Mental Health Signal Detector, Los Angeles, CA Aug – Dec 2025

- Assembled a RAG-enabled NLP content classification pipeline over 10K+ Reddit posts, improving intent classification by 10%; fine-tuned and benchmarked DistilRoBERTa, BERT, and MiniLM with LoRA.